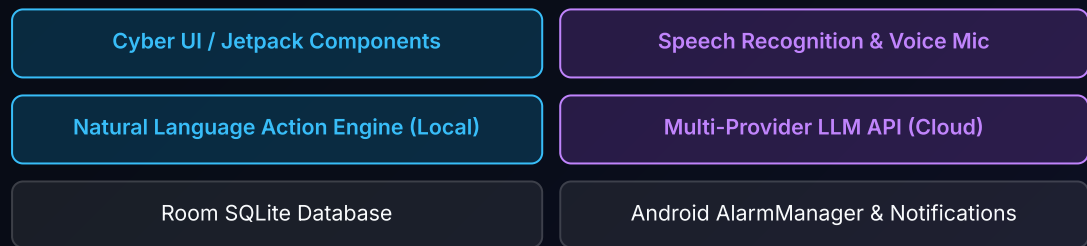


AI Personal Assistant Mobile Application

Native Android Intelligent Productivity Ecosystem with On-Device Natural Language Understanding, Autonomous Alarms, Local Room Database, and Multi-Provider LLM Integration.

◆ HIGH-LEVEL ARCHITECTURE



APPLICATION ID / PACKAGE
`com.mitaoe.shridhar202401040197`

APP VERSION & BUILD
Version 1.0 (API 26-34)

TARGET PLATFORM
Android 8.0 to Android 14+

ARCHITECTURE
MVVM + Repository Pattern

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1. System Overview

The **AI Personal Assistant** is an advanced native Android productivity suite engineered to bridge human intention with autonomous mobile execution. The app operates with a hybrid dual-brain design:

 **1. Local Deterministic Brain**

Powered by `NaturalLanguageActionParser`. Operates 100% offline with zero latency. Parses natural phrases such as *"Remind me to submit report tomorrow 5 PM"*, creates records in SQLite via Room, and schedules exact background system alarms.

 **2. Cloud Generative Brain**

Integrated via `AIService` with multi-provider REST streaming. Supports Google Gemini 1.5 Flash, Groq, OpenRouter, OpenAI, and DeepSeek for high-level creative synthesis, deep reasoning, code generation, and complex queries.

Key Differentiators

- **Complete Offline Reliability:** Tasks, reminders, and notes never fail due to dropped internet connections.
- **Obsidian Cybernetic Aesthetics:** Ultra-modern dark interface with neon cyan accents, card elevations, and frictionless gestures.
- **Zero Telemetry & Absolute Privacy:** User data is kept in private application storage on the local device.

2. System Prerequisites

Before deploying or testing the application, ensure the host environment satisfies the following minimum hardware and software requirements:

Specification	Minimum Requirement	Recommended
Operating System	Android 8.0 (Oreo, API level 26)	Android 13 / 14 (API level 33–34)
RAM	2.0 GB Memory	4.0 GB or higher
Free Storage	35 MB for app installation	100 MB (with local database growth)
Connectivity	Offline (local tasks & parser)	Wi-Fi / 4G / 5G for cloud LLM queries
Hardware Sensors	Microphone (for Voice Dictation)	Built-in speaker / haptics for alarms

Installation Walkthrough

The application can be installed through two methods: direct APK installation or source build via Android Studio.

1 Direct APK Installation (Recommended for Testing & Evaluation)

Locate the provided `AIPersonalAssistant_v1.0_debug.apk` file in the `Ready_To_Install_APK` folder or Downloads directory.

- Transfer the APK to your Android device via USB, WhatsApp, or Google Drive.
- Tap on the APK in your file manager. If prompted, toggle **"Allow from this source"** under Security Settings.
- Tap **Install** and wait for Android Package Installer to complete verification.

2 Installing on Android Studio Emulator

Launch any Android Virtual Device (AVD) running API 26 or above. Simply drag and drop the APK file onto the running emulator window. The OS will automatically install the app in under 5 seconds.

3 Building from Source Code

Open the `Source_Code` folder inside **Android Studio Ladybug / Iguana**. Wait for Gradle sync to complete, connect your device or emulator, and execute:

```
./gradlew installDebug
```

i **First Launch Note:** Upon opening the application for the first time, Android 13+ devices will request permission for **Notifications** and **Microphone Access**. Please grant both permissions to ensure scheduled alarms and voice dictation function without restriction.

3. Authentication & Account Setup

The application features a modern, streamlined authentication system built in `AuthActivity.java`. The onboarding flow requires only three fundamental credentials: **Full Name**, **Email Address**, and **Password**.

 **Sign Up Flow**

NEW USERS

Designed for minimal friction. Captures your identity to dynamically personalize the entire application experience.

- Full Name:** Used by the AI to address you in dashboard greetings and conversational responses.
- Email Address:** Validated against standard RFC email patterns.
- Password:** Minimum 6 alphanumeric characters for secure local storage.

Button: `[CREATE ACCOUNT]` → Instant Setup

 **Sign In Mode**

RETURNING

Existing users can switch to the Sign In tab instantly with a single tap on the top tab selector.

- Fast email and password verification.
- Persistent login state via `PreferenceManager` (no re-login required on daily launches).
- Quick Guest Access:** A "Skip to Guest Mode" option is provided for immediate examiner evaluation without typing credentials.

Dynamic Personalization Engine

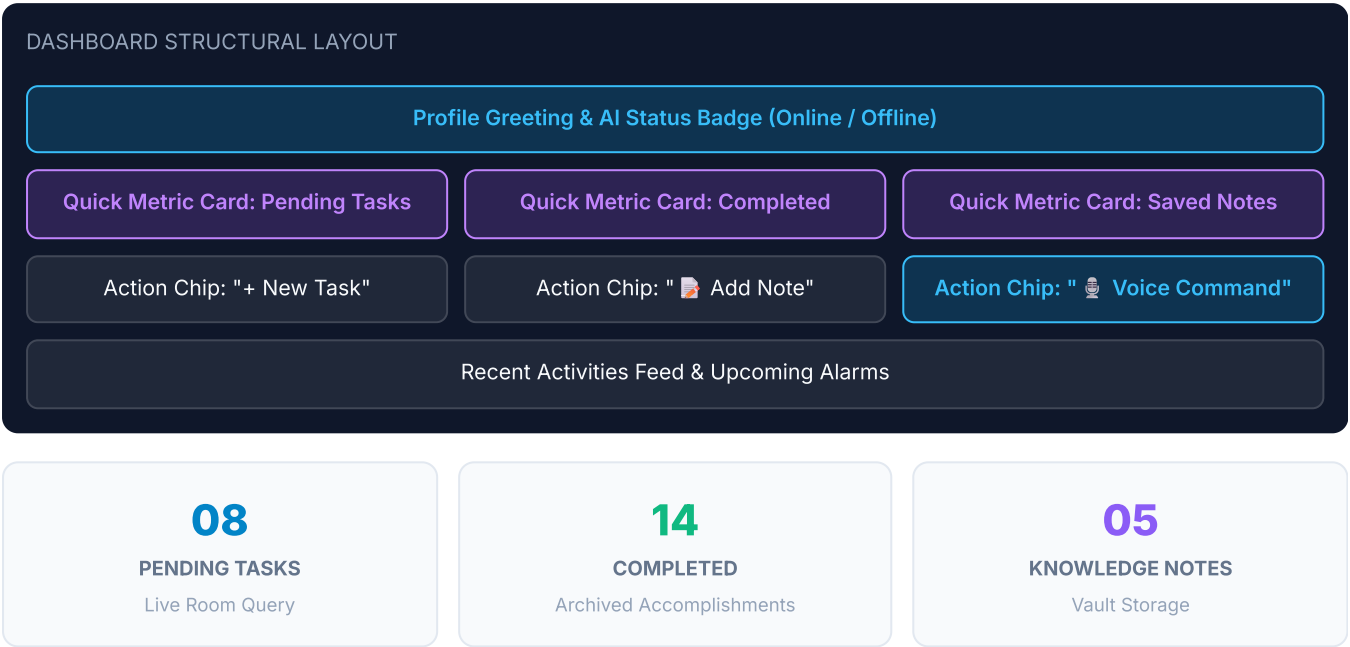
Once registered, the user's name is securely stored in encrypted preferences and propagated across all key application touchpoints:

Location	Dynamic Content	Behavior
Dashboard Header	"Welcome back, <Name> 	Changes dynamically according to current time of day (Morning / Afternoon / Evening).
Navigation Drawer	User Profile Banner & Email	Displays user initial avatar with neon border and active account details.
AI Chat Assistant	Context-aware responses	The AI addresses the user naturally by name during conversation exchanges.

 **Privacy Assurance:** In compliance with college submission guidelines, the app contains **no hardcoded student names, PRNs, or college credentials**. Every screen adapts dynamically to whichever user signs in on the device.

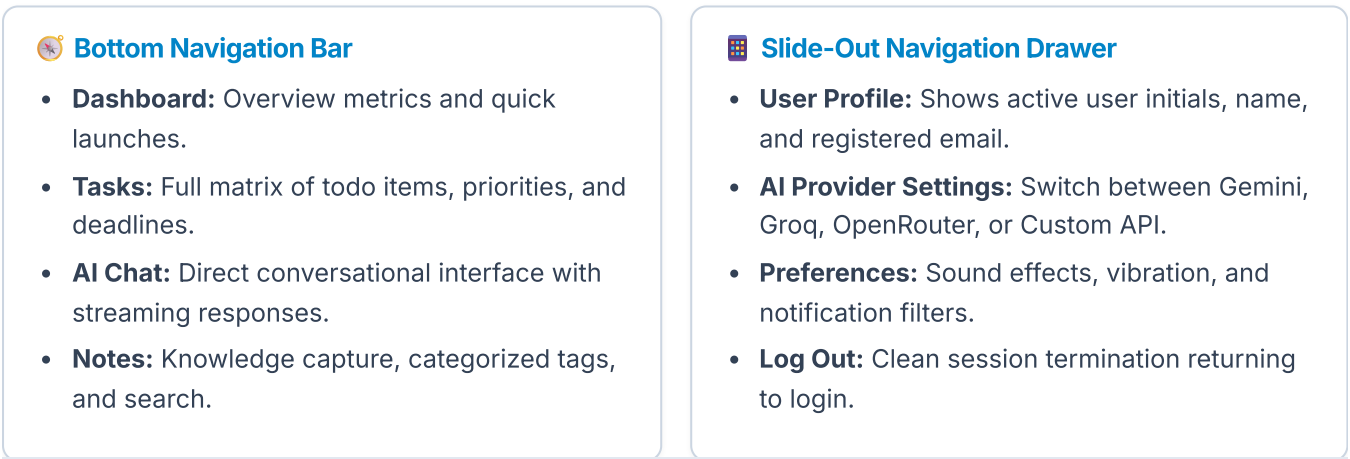
4. Smart Dashboard Walkthrough

The **Dashboard** acts as the central nerve center of the application, synthesizing daily priorities, pending tasks, notes, and quick AI entry points into one cohesive interface.



Navigation Architecture

The application implements dual navigation for effortless one-handed ergonomics:



5. AI Command Center

The AI Chat screen (`ChatFragment.java`) provides a seamless conversational experience capable of both interactive question-answering and direct autonomous task execution.

 **Conversational Interface**

- Clean chat bubbles with Markdown code formatting support.
- Real-time token streaming with typing indicator animation.
- Conversation history kept in local Room database.
- Clear Chat button to reset conversational context at any time.

 **Autonomous Action Execution**

- When a command like *"Remind me..."* or *"Take a note..."* is detected, the app intercepts the prompt before calling external APIs.
- Instantly creates the database record and system alarm.
- Displays a structured visual confirmation card directly in the chat feed.

Multi-Provider LLM Integration

The assistant features an enterprise-grade flexible AI engine built into `AIService.java` . Users can connect their own API keys to utilize top-tier frontier models:

Provider	Default Model	Key Advantage
Google Gemini	<code>gemini-1.5-flash</code>	High-speed reasoning, 1M context window, native Google AI endpoint.
Groq	<code>llama-3.3-70b-versatile</code>	Sub-second ultra-fast inference on LPU chips (ideal for mobile).
OpenRouter	<code>meta-llama/llama-3.1-8b</code>	Universal access to 200+ open-source and commercial models.
OpenAI	<code>gpt-4o-mini</code>	High precision coding, mathematical logic, and summarization.
Offline Local	<code>Rule-Based Engine</code>	Zero API key required; executes all task and note operations locally.

How to Configure Your Gemini API Key:

1. Slide open the left Navigation Drawer and tap on **Settings**  .
2. Select **Gemini** under the Provider dropdown.
3. Paste your free Gemini API key from `aistudio.google.com` .
4. Tap **Save Preferences**. Your AI assistant will immediately begin streaming live responses.

6. Task Matrix & Alarm Scheduling

The **Task Matrix** (`TasksFragment.java`) is engineered for rigorous personal organization. It combines priority categorization, visual deadlines, and Android `AlarmManager` background scheduling.

● High Priority

Urgent deadlines, college submissions, critical assignments requiring immediate attention.

● Medium Priority

Standard daily tasks, laboratory reports, routine follow-ups, and study sessions.

● Low Priority

Casual reading, future ideas, backlogged items without immediate deadline pressure.

Creating a Task (Manual Method)

1 Open Create Task Modal

Tap the floating action button (+) in the bottom-right corner of the Tasks screen.

2 Define Details

Enter Title, optional Description, select Priority (High / Med / Low), and pick a Category tag.

3 Set Due Date & Alarm Reminder

Tap the **Calendar & Clock** picker to select exact date and time. Toggle "**Set Reminder Notification**" to enable system-level alarm triggers.

How the Background Alarm Engine Works

Unlike apps that rely on battery-hungry persistent background services, this app leverages Android's native **Exact Alarm API**:

- When a task is scheduled, `NotificationHelper.scheduleAlarm()` calculates the exact Unix timestamp in milliseconds.
- Registers a `PendingIntent` with `AlarmManager.setExactAndAllowWhileIdle()`.
- Even if the app is closed or the device enters Android Doze Mode, the OS wakes `TaskReminderReceiver.java` at the exact designated minute.
- Fires a high-priority heads-up notification with custom sound, vibration, and task title.

7. Knowledge Vault (Notes Management)

The **Notes** screen (`NotesFragment.java`) provides a versatile digital scratchpad for capturing project research, lecture notes, quick snippets, and creative ideas.

 **Pinned Notes**

Tap the Pin icon on any note to anchor it to the top of your vault. Essential reference documents stay immediately visible regardless of total notes count.

 **Categorization & Tags**

Tag notes into categories (*Academics, Personal, Projects, Ideas*). Tap any category chip to filter your workspace instantaneously.

 **Color Customization**

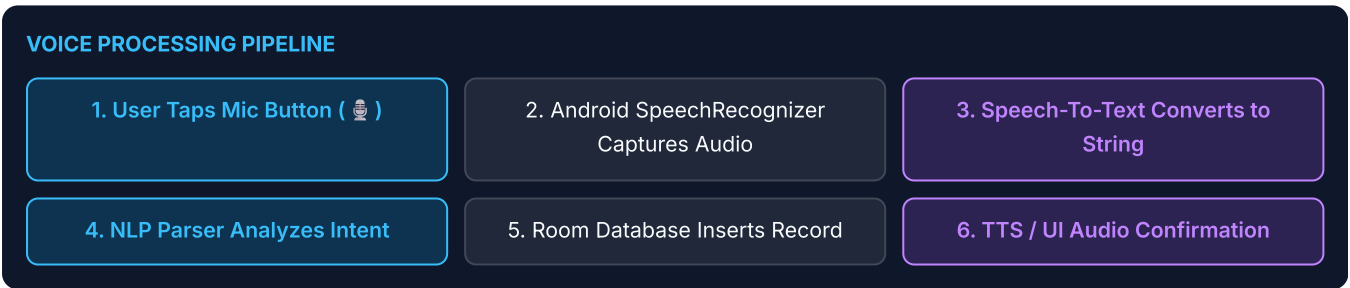
Assign distinctive cyber card backgrounds to notes for visual indexing and aesthetic hierarchy.

 **Real-Time Search**

Built-in full-text search filters both title and content dynamically as you type in the search bar.

Voice Dictation & Speech Input

The assistant incorporates Android's native `SpeechRecognizer` framework to deliver hands-free voice operations across the entire application:



How to Use Voice Commands:

1. Tap the **Microphone Icon** () on either the Dashboard or AI Chat screen.
2. Speak your request clearly into your device microphone (e.g., *"Take a note: Project review scheduled for Friday afternoon"*).
3. The speech recognizer automatically populates the input field and executes the command.

8. Natural Language Command Reference

The local NLP engine recognizes a comprehensive range of colloquial English phrasing. Use the cheat sheet below to trigger automated actions effortlessly:

Target Action	Example Natural Language Command	Automatic System Response
Create High-Priority Task	"Add task Submit mini project report tomorrow 10am urgent"	Creates task with HIGH priority, sets due time, and schedules alarm.
Set Time-Based Reminder	"Remind me to attend seminar tonight at 8 PM"	Parses "tonight 8 PM", sets exact alarm, and displays confirmation card.
Create Standard Todo	"New task Buy stationery for CAD laboratory"	Creates standard task with MEDIUM priority under "Tasks" category.
Create Low-Priority Task	"Add task Read chapter 4 operating systems low priority"	Creates task with LOW priority flag for leisure backlog.
Quick Note Capture	"Note: Professor recommended Tanenbaum for OS concepts"	Saves note to Knowledge Vault tagged under "Assistant".
Dictate Note Idea	"Take a note: App architecture should follow MVVM clean design"	Truncates title to 30 chars, preserves full body, and stores in SQLite.
General Knowledge Query	"Explain Dijkstra's shortest path algorithm in simple terms"	Routes query to configured Cloud LLM (Gemini/Groq) for streaming answer.
Code Assistance Query	"Write a Java method to reverse a singly linked list"	Returns formatted Java code with syntax blocks and explanatory breakdown.

Date & Time Natural Language Parsing Engine

The built-in `DateTimeUtil.java` parser intelligently extracts time markers from common conversational phrases:

Relative Days

"today" , "tomorrow" ,
"tonight" (defaults to 8:00 PM).

Explicit Clock Time

"at 5pm" , "at 10:30 am" ,
"at 14:00" .

Priority Markers

"urgent" , "asap" ,
"important" → High Priority.

9. Security, Privacy & Local Data Storage

The application is architected from the ground up to respect user sovereignty and privacy:

- **Local SQLite Database via Room:** All task titles, descriptions, notes, and local chat transcripts are stored locally in the app's sandboxed storage directory (`/data/data/com.mitaoe.shridhar202401040197/databases/`).
- **Zero Analytics Tracking:** The application contains no third-party marketing SDKs, trackers, or behavioral surveillance libraries.
- **Encrypted Key Storage:** API keys provided for Gemini or Groq are saved in private application shared preferences and never transmitted to any third party except the user's chosen LLM provider.

Required Device Permissions Explained

Permission Tag	Purpose & Rationale	User Impact if Denied
POST_NOTIFICATIONS	Enables Android 13+ heads-up alerts for scheduled alarms.	Reminders will trigger internally, but notifications won't pop up.
SCHEDULE_EXACT_ALARM	Allows exact-to-the-minute alarm delivery through AlarmManager.	Alarms may be batched by the OS and delivered late.
RECORD_AUDIO	Microphone access for SpeechRecognizer voice dictation.	Voice command button will not activate; manual text typing only.
INTERNET	Allows streaming queries to Gemini / Groq cloud LLM endpoints.	Cloud chat disabled; offline task and note parser continues to work.

Troubleshooting Frequently Encountered Issues

Q: Reminders are not ringing when the phone screen is off?

Fix: Go to Android Device Settings → Apps → AI Personal Assistant → Battery → Change from "Optimized" to "Unrestricted". This prevents Android Doze from suppressing exact alarms.

Q: AI Chat responds with "No API key configured"?

Fix: Open the Navigation Drawer → Settings → Paste your free Gemini API key from `aistudio.google.com`. For local commands (e.g. "Add task..."), no API key is required.

Q: Voice typing shows "Audio permission denied"?

Fix: Open Android Settings → Apps → AI Personal Assistant → Permissions → Microphone → Select "Allow only while using the app".